

MUHAMMAD HUSSAIN KHAN LODHI

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Summary

Software engineer with a strong foundation in data structures, object-oriented design, and database systems, and particular depth in full-stack MERN development (React, Node.js, Express, MongoDB). Has designed multi-service systems, including a deterministic recommendation engine and a role-based access-control platform, with production experience across Java and C/C++ as well. IBM-certified in full-stack JavaScript development; BSCS graduate, CGPA 3.7/4.0.

Experience

NextSkill Institute

Jun 2026 - Sep 2026

Full Stack Development Trainee

Karachi, Pakistan

- Completed a NAVTTC-affiliated Full Stack Development training program with an IBM-aligned curriculum, progressing from HTML5/CSS3/JavaScript fundamentals to React-based multi-page and e-commerce applications
- Built and version-controlled a series of front-end projects, including a personal portfolio, an expense tracker, and a multi-page e-commerce site, using Git/GitHub for iterative submission and review

10Pearls - 10Shine Program

[Certificate](#) | Apr 2026 - May 2026

MERN Stack Development Intern

Remote, Karachi

- Designed and implemented the data model and REST API for a full-stack Notes application (React, Node.js, Express, MySQL), including JWT authentication, email verification, and full CRUD (add, update, delete, search)
- Authored unit/integration tests (Mocha, Chai, NYC coverage) and resolved issues flagged by SonarQube static analysis; collaborated via Git/GitHub with pull requests and code review

Self-Directed Learning & Development

Sep 2022 - Present

Independent Study

Karachi, Pakistan

- Designed, built, and deployed 10+ full-stack MERN applications (React, Node.js, MongoDB/MySQL) independently, applying data structures, OOP, and database design coursework to real projects
- Practiced collaborative software workflows (Git/GitHub) and iterative, test-driven development

Projects

Smart Complaint Management System (MERN Stack)

[GitHub](#) | [Live Demo](#)

- Designed a role-based access-control system and a two-stage state machine, covering account approval and complaint lifecycle (Pending, In Progress, Resolved/Rejected), enforced across 15 REST endpoints
- Built the data model and API layer (React, Node.js, Express, MongoDB/Mongoose) with JWT auth, bcrypt hashing, and admin-only route guards; deployed frontend and backend as independent Vercel projects

Notes Management Platform (Full-Stack)

[GitHub](#) | [Live Demo](#)

- Architected a full-stack notes application (React, Node.js, Express, MySQL) with user registration, login, JWT-based authentication, and email verification workflows
- Engineered full CRUD operations with title/content search, automated endpoint testing, and independent Vercel deployments with cloud MySQL (TiDB Cloud) integration

BuildVolt: PC-Build Recommendation SaaS (Final Year Project)

[GitHub](#) | [Live Demo](#)

- Designed a deterministic compatibility engine (Node.js, Express, Turso/libSQL) that infers CPU/RAM/PSU specs from catalog text and assembles budget-capped, safety-checked PC builds across 1,200+ lines of logic
- Built a WordPress/WooCommerce integration and web dashboard for catalog sync, CSV/Excel import, analytics, and JWT-secured plugin endpoints with Resend-powered onboarding/support-ticket emails

Education

Sindh Madressatul Islam University

Sep 2022 - Jun 2026

Bachelor of Science in Computer Science; CGPA: 3.7/4.0

Karachi, Pakistan

- **Relevant Coursework:** Data Structures, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks, Operating Systems, Mobile App Development

Certifications

IBM Full-Stack JavaScript Developer (Professional Certificate)

[Verify](#) | Sep 2026

IBM, Coursera

Online

Modern Web & App Development

Mar 2026 - Present

Saylani Mass IT Training (SMIT)

Karachi, Pakistan

Technical Skills

Languages: JavaScript, TypeScript, JSX, HTML5, CSS3, Python, Java, C++, C, SQL, PHP, Shell

Core CS: Data Structures, Object-Oriented Programming, REST API Design, Database Design, Unit Testing

Frontend: React.js, React DOM, Next.js, React Router, Tailwind CSS, Context API, Redux Toolkit, React Redux, Vite, Framer Motion

Backend / APIs: Node.js, Express.js, Socket.IO, bcrypt, JWT, CORS, dotenv, Multer, Node.js http, fs/promises, path, JSON

Databases / ORM: MongoDB, MongoDB Atlas, Mongoose, PostgreSQL, pg, Sequelize, MySQL, Turso (libSQL), SQL Server

Developer Tools: npm, Git, GitHub, GitHub Actions, Docker, Dockerfile, Nodemon, ESLint, React Hooks ESLint, Postman, SonarQube, Mocha/Chai, VS Code, NetBeans

Deployment / Cloud: Vercel, GitHub Pages